Midterm-test prepare / All theory

Chap 1

1. Network Transmission Quality

- Bandwidth, Congestion, Delay, and Jitter
- The typical 3 congestion points are aggregation, speed mismatch, and LAN to WAN.

2. Main purpose of QoS

- Prioritize important traffic (Exp: VOIP)
- Reduce delay for time-sensitive applications
- Minimize packet loss during network congestion

3. Jitter

- is the variation of delay of received packet
- do not arrive at consistent time intervals

4. Queuing Delay Vs Serializing Delay

	Queuing Delay	Serializing Delay
Define	Times a packet wants in a queue before transmission	Time required to place all packet bits into the transmission link
cause	network congestion or busy interface	packet size and link bandwidth

5. Four types of Queuing Algorithms

First in First Out (FIFO)

- queuing buffers and forwards packets in the order of their arrival.

Advantage of (FIFO)

- simple and easy to implement

Disadvantages of (FIFO)

- no QoS support

Weighted Fair Queuing (WFQ)
- Automatically classifies traffic into different flows
- Assigs bandwidth fairy based on weight

Advantage of (WFQ)
- Fair bandwidth allocation, prevents starvation support.
Disadvantages of (WFQ)
- More Complex than FIFO

Class Based Weighted Fair Queuing (CBWFQ)
- Traffic is manually classified into classes

Advantage of (CBWFQ)
- Better traffic control
Disadvantages of (CBWFQ)
- More complex to configured

Low Latency Queuing (LLQ)
- N/A

Advantage of (LLQ)
- Minimize delay and jitter

Disadvantages of (LLQ)
- More complex to configured

6. Main QoS Models

Best-effort model
Description
- Not an implementation as QoS is not explicitly configured.
- All packets were treated the same way.

Benefits/advantages:
- Most scalable
- no special QoS mechanisms required.

Drawbacks/disadvantages
- No guarantees of delivery

Integrated services (IntServ)
Description
- Provides very high QoS to IP packets with guaranteed delivery.

Protocol used: Uses RSVP

Benefits/advantages
- Explicit end-to-end resource admission control
- signaling of dynamic port numbers.

Drawbacks/disadvantages
- flow-based approach not scalable to large implementations such as the internet.

Differentiated services (DiffServ)
Description
- Provides high scalability and flexibility in implementing QoS.
- Not an end-to-end QoS strategy

Benefits/advantages
- Highly scalable
- provides many different levels of quality.

Drawbacks/disadvantages
- No absolute guarantee of service quality

7. Three QoS Tools Categories
- Classification and Marking tools
- Congestion Avoidance tools
- Congestion management tools

8. DSCP — 3 Main Categories

Category	DSCP Value
Best Effort (BE)	0
Expedited Forwarding (EF)	46 (binary 101110)
Assured Forwarding (AF)	AFxy formula

9.

A. Traffic Characteristics（完全没读到，这个很重要）

这题很可能会考一个**比较表**（Voice vs Video vs Data），务必背熟数字：

	Voice	Video	Data
特性	Smooth, predictable, benign	Bursty, unpredictable, greedy	Smooth/bursty, drop insensitive
Latency	< 150ms	< 200-400ms	不敏感
Jitter	< 30ms	< 30-50ms	不敏感
Loss	< 1%	< 0.1-1%	drop insensitive (TCP会重传)
Bandwidth	30-128 Kbps	384 Kbps - 20 Mbps	视情况
Protocol	UDP, RTP port 16384-32767	UDP, RTSP port 554	TCP

Chap 2

1. Telnet and SSH

	Telnet	SSH
Port number	Port num 23	Port num 22
Encryption	No encryption. Data is sent in plaintext and can be easily intercepted. Security level low	All transmitted data is encrypted, including login credentials. Security level high
Authentication	Basic username/password only. Credentials are transmitted in plaintext.	Supports encrypted username/password and stronger methods like public key authentication.

2. File Transfer Protocol (FTP), FTPS, SFTP

	FTP	FTPS	SFTP
Port number	TCP port 21 (control) and TCP port 20 (data)	Explicit: port 21 Implicit: port 990	TCP port 22
security	Not secure. No encryption protects data during transfer.	Secure. Adds SSL/TLS encryption to standard FTP.	Most secure. Built on SSH with full encryption.
Authentication	Basic username/password only.	Username/password + TLS/SSL certificate for server authentication.	Username/password + SSH key authentication.

Chap 3 - Simple Network Management Protocol (SNMP)

SNMP function

- monitor network performance and device status

SNMP system consists of three elements

SNMP manager

- Monitors and control network devices

F - send SNMP requests and receive responses

SNMP management agents

- Software on managed device that collects & reports info to NMS

F - Responds to manager requests and sends traps / alerts

Management Information Base (MIB)

- Database containing information about managed device

F - Store information that the manager and agent access

UDP Port Numbers

- SNMP manager polls agents on UDP port 161
- SNMP agents send traps to the manager on UDP port 162

SNMP advantages

- Easy network monitoring
- Detects problems quickly.

SNMP operation

operation	description
get	SNMP manager retrieves info from an agent
set	SNMP manager changes configuration on an agent
trap	The agent sends an alert to the manager without being asked.
get-request	Retrieves a value from a specific variable
get-next-request	Retrieves the next variable in the MIB table
get-bulk-request	Retrieves multiple value in one request
get-response	Agent replies to GET, GET-NEXT, or SET requests
set-request	Stores or changes a value on the managed device.

SNMP Versions and Security

- SNMPv1: The earliest, with basic community-based security.
- SNMPv2c: Improved performance but maintained similar security.
- SNMPv3: Offers robust security features, including user-based authentication, encryption, and access control.

Chap 4 - Syslog (system log)

Syslog define
- Standard protocol for generating, storing, and processing event (log) messages.

Port Numbers
Syslog uses UDP port 514 to send messages

Syslog standard addresses three primary components
Event Message Format
- Define how log messages are organized.

Event Message Transmission
- Sends log messages across the network

Event Message Handling
- Syslog utility creates, receives, analyzes, and stores event messages.

Syslog Logging Service — 3 Primary Functions
- The ability to gather logging information for monitoring and troubleshooting
- The ability to select the type of logging information that is captured
- The ability to specify the destinations of captured syslog messages

Syslog Operation — Message Destinations
Syslog messages can be sent to different destinations:
- Logging buffer
- Console line
- Terminal line
- Syslog server

Severity Level
- The smaller the numerical level, the more critical the message

Facility
- Facility options are specific to the networking device. Exp: IP

Syslog Message Format / Severity Level

Severity Name	Severity Level	Explanation
Emergency	Level 0	System Unusable
Alert	Level 1	Immediate Action Needed
Critical	Level 2	Critical Condition
Error	Level 3	Error Condition
Warning	Level 4	Warning Condition
Notification	Level 5	Normal, but Significant Condition
Informational	Level 6	Informational Message
Debugging	Level 7	Debugging Message

Identify the error in the question (Maybe will have)

Syslog

Configure Syslog Timestamp

By default, log messages are not timestamped. Log messages should be timestamped so that when they are sent to another destination, such as a Syslog server, there is record of when the message was generated. Use the command **service timestamps log datetime** to force logged events to display the date and time.

```
R1# configure terminal
R1(config)# interface g0/0/0
R1(config-if)# shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down
R1(config-if)# exit
R1(config)# service timestamps log datetime
R1(config)# interface g0/0/0
R1(config-if)# no shutdown
*Mar 1 11:52:42: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
*Mar 1 11:52:45: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Mar 1 11:52:46: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0,
changed state to up
R1(config-if)#
```

Message Format

F. Message Format 公式（很重要，考试超爱考"identify the parts"）

Default Cisco IOS syslog message format:

> %facility-severity-MNEMONIC: description

Example (memorize this exact one, it's from the slide):

> `%LINK-3-UPDOWN: Interface Port-channel1, changed state to up`

Breakdown:

- **Facility** = LINK
- **Severity** = 3 (Error)
- **MNEMONIC** = UPDOWN
- **Description** = "Interface Port-channel1, changed state to up"

G. Configure Syslog Timestamp（你写了截图，这里补充command细节）

- By default, log messages are **NOT timestamped**
- Command to enable timestamp:
  > `service timestamps log datetime`
- Purpose: so when messages are sent to a destination (e.g. syslog server), there's a **record of when the event happened**

Before command (no timestamp):

```
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to administratively
down
```

After command (with timestamp):

```
*Mar 1 11:52:42: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to
down
```

Command sequence to remember:

```
R1(config)# service timestamps log datetime
```

service timestamps log datetime